WORKED SOLUTION

STEP III 1991, Question 6

Pure • Hyperbolic Linear Transformations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q6

This question has a full guided treatment in *Beyond STEP*, Section 7.6, *Hyperbolic Rotations and Relativity*: https://beyondstep.keithmcnulty.org/ch07/hyperbolic-rotations.html

The key idea

The matrix behaves like a rotation matrix, but with hyperbolic functions in place of circular ones, and the identities $\cosh^2 u - \sinh^2 u = 1$ and the addition formulae for cosh and sinh do all the work. Writing a direction of gradient $\tanh v$ as the vector $(\cosh v, \sinh v)$ makes the effect of T on gradients obvious: it simply adds u to the "hyperbolic angle" v. For the angle condition, we express the tangent of the angle between two lines in terms of v_1 and v_2 and see what has to stay fixed.

Solution

Throughout, write $c = \cosh u$ and $s = \sinh u$, so that $c^2 - s^2 = 1$, and $s > 0$ because $u > 0$. The transformation is

$$x' = cx + sy, \qquad y' = sx + cy.$$

The hyperbola

For any point,

$$\begin{aligned} x'^2 - y'^2 &= (cx + sy)^2 - (sx + cy)^2 \\ &= (c^2 - s^2)x^2 + 2csxy - 2csxy - (c^2 - s^2)y^2 \\ &= x^2 - y^2. \end{aligned}$$

So T preserves $x^2 - y^2$, and every point of $x^2 - y^2 = 1$ is sent to a point of $x^2 - y^2 = 1$. The curve is covered completely, not just mapped into itself: the matrix with $-u$ in place of u is the inverse of T (multiply out and use $c^2 - s^2 = 1$), and it also preserves

$x^2 - y^2$. So every point of the hyperbola is the image of a point of the hyperbola. The curve transforms into itself.

(In fact each branch goes to itself: if $x \geq 1$ then $|y| < x$, so $x' = cx + sy > (c - s)x = \mathrm{e}^{-u}x > 0$.)

Invariant lines through the origin

Since T is linear, it sends the origin to itself and the line through the origin in direction $\mathbf{d}$ to the line through the origin in direction $T\mathbf{d}$. So we need $T\mathbf{d}$ parallel to $\mathbf{d}$, that is, $\mathbf{d}$ an eigenvector. Try $\mathbf{d} = (1, m)$:

$$T\begin{pmatrix}1\\m\end{pmatrix} = \begin{pmatrix}c+sm\\s+cm\end{pmatrix},$$

which is parallel to $(1, m)$ exactly when $s + cm = m(c + sm)$, that is, $s(1 - m^2) = 0$. Since $s \neq 0$, $m = \pm 1$. (The vertical direction $(0, 1)$ goes to (s, c), which is not vertical, so it gives nothing new.) The two lines are

$$\boxed{y = x \quad \text{and} \quad y = -x}$$

and indeed $T(1, 1) = \mathrm{e}^u(1, 1)$ and $T(1, -1) = \mathrm{e}^{-u}(1, -1)$. These are the asymptotes of the hyperbola, which fits with the hyperbola mapping to itself.

Gradients: $v' = v + u$

Take a line through a point $\mathbf{p}$ with direction $\mathbf{d}$, so its points are $\mathbf{p} + t\mathbf{d}$. By linearity its image is $T\mathbf{p} + tT\mathbf{d}$: a line through $T\mathbf{p}$ with direction $T\mathbf{d}$. So the position of the line is irrelevant and only the direction matters.

A line of gradient $\tanh v$ has direction $(1, \tanh v)$, or, multiplying by $\cosh v > 0$, direction $(\cosh v, \sinh v)$. Then

$$T\begin{pmatrix}\cosh v\\\sinh v\end{pmatrix} = \begin{pmatrix}\cosh u\cosh v + \sinh u\sinh v\\\sinh u\cosh v + \cosh u\sinh v\end{pmatrix} = \begin{pmatrix}\cosh(u+v)\\\sinh(u+v)\end{pmatrix}$$

by the addition formulae. The image line therefore has gradient $\tanh(v + u)$. Since $\tanh$ is one-to-one, $\tanh v' = \tanh(v + u)$ gives

$$v' = v + u,$$

as required. (Only lines with gradient strictly between -1 and 1 can be written this way, which is why the question uses $\tanh v$.)

Equal angles

If two lines have gradients m_1 and m_2 and $1 + m_1m_2 \neq 0$, the acute angle θ between them satisfies

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1m_2}\right|.$$

Here $|m_1|, |m_2| < 1$, so $1 + m_1 m_2 > 0$: the lines are never perpendicular, $\theta \in [0, \frac{\pi}{2})$, and θ is fixed by $\tan\theta$.

Ask yourself The formula looks like $\tanh(v_1 - v_2)$, but the sign in the denominator is wrong. What does it simplify to instead?

Put $m_i = \tanh v_i$ and multiply the top and bottom by $\cosh v_1 \cosh v_2 > 0$:

$$\tan\theta = \frac{|\sinh v_1 \cosh v_2 - \cosh v_1 \sinh v_2|}{\cosh v_1 \cosh v_2 + \sinh v_1 \sinh v_2} = \frac{|\sinh(v_1 - v_2)|}{\cosh(v_1 + v_2)}.$$

The transformed lines have $v_i' = v_i + u$. Their difference is unchanged, but their sum goes up by $2u$, so the angle θ' between them satisfies

$$\tan\theta' = \frac{|\sinh(v_1 - v_2)|}{\cosh(v_1 + v_2 + 2u)}.$$

For ℓ_1 and ℓ_2 to intersect at all we need $v_1 \neq v_2$, so the numerator is not zero. Hence

$$\theta = \theta' \iff \cosh(v_1 + v_2) = \cosh(v_1 + v_2 + 2u).$$

Since $\cosh a = \cosh b$ exactly when $a = \pm b$, this means either $v_1 + v_2 = v_1 + v_2 + 2u$, which is impossible as $u > 0$, or $v_1 + v_2 = -(v_1 + v_2 + 2u)$. So the necessary and sufficient condition is

$$\boxed{v_1 + v_2 = -u}$$

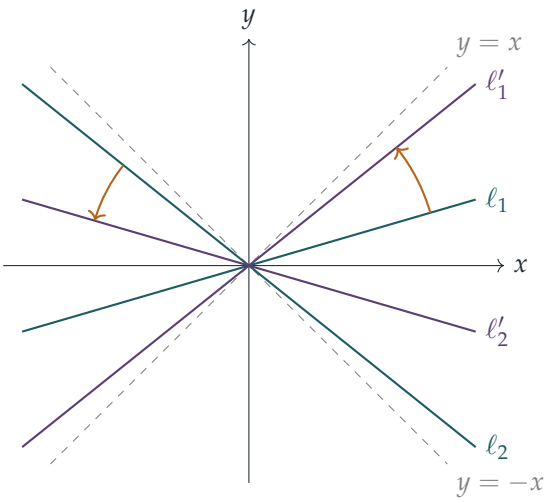
or equivalently $v_1' + v_2' = u$.

The picture when the lines meet at the origin

When $v_1 + v_2 = -u$ we have

$$v_1' = v_1 + u = -v_2, \qquad v_2' = v_2 + u = -v_1.$$

So ℓ_1' has gradient $-\tanh v_2$: it is the reflection of ℓ_2 in the x-axis. Likewise ℓ_2' is the reflection of ℓ_1. If the lines pass through the origin, the transformed pair is just the mirror image of the original pair in the x-axis, and that is why the angle between them is the same. The diagram uses $u = 0.8$, $v_1 = 0.3$, $v_2 = -1.1$: the gradients are about 0.29 and -0.80 for ℓ_1, ℓ_2, and 0.80 and -0.29 for ℓ_1', ℓ_2'. Each line turns towards $y = x$, the invariant line that T stretches.



Teal lines are ℓ_1, ℓ_2, plum lines their images; the amber arrows show $\ell_1 \to \ell_1'$ and $\ell_2 \to \ell_2'$. The plum pair is the reflection of the teal pair in the x-axis.

Checks

- With $u = 0.8$, $v_1 = 0.3$, $v_2 = -1.1$ the angle between ℓ_1 and ℓ_2 and between their images both come to 54.92° numerically. Changing v_2 slightly makes the two angles differ.
- As $u \to 0$, T becomes the identity, and the condition becomes $v_1 + v_2 = 0$: two lines symmetric about the x-axis, which an identity map trivially keeps symmetric.
- $T(1,1) = \mathrm{e}^u(1,1)$ is easy to check by hand, since $\cosh u + \sinh u = \mathrm{e}^u$.

Common slips

- Showing only that the hyperbola maps *into* itself; the inverse transformation shows nothing is missed.
- Writing the angle formula as $\tanh(v_1 - v_2)$; the $+$ sign in $1 + m_1 m_2$ gives $\cosh(v_1 + v_2)$ in the denominator instead, and it is the *sum* that changes under T.
- Forgetting the root $v_1 + v_2 = v_1 + v_2 + 2u$ of $\cosh a = \cosh b$ and not saying why it is impossible.

Beyond STEP

This matrix is a Lorentz boost from special relativity, with x and y playing the roles of time and space (in units where the speed of light is 1). The quantity $x^2 - y^2$ is the invariant interval, $\tanh v$ is a velocity, and v is its *rapidity*. The result $v' = v + u$ says that

rapidities add, which is the relativistic velocity-addition formula in disguise. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops special relativity this way. For the eigenvector view of the invariant lines, see Strang, *Introduction to Linear Algebra*.

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